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align(center)[ text(fill: black90, size: 11pt, weight: "bold") [title] linebreak() text(fill: black70, size: 8pt)
[subtitle] linebreak() text(fill: black70, size: 6.8pt) [Five native noisy fluorescence fields. Quantitative
summary combines analytical white-noise gain with a dark-background gradient-stability proxy.] ]
v(8pt) image-grid(data, [Gradient magnitude, native fluorescence images], "magnitude_path", show-
input: true)
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align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[Orientation maps. Hue encodes edge angle modulo  $\pi$  and brightness follows normalized
magnitude.] ] v(8pt) image-grid(data, [Gradient orientation, native fluorescence images],
"orientation_path", show-input: false)
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align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[Analytical and empirical stability summary. The empirical proxy is the median absolute deviation of
gradient magnitude on the darkest percentile pixels.] ] v(8pt) grid( columns: 2, column-gutter: 10pt,
[ bar-chart(data, "white_noise_gain", [Analytical white-noise gain], [log10 WNG], log-scale: true) ],
[ bar-chart(data, "background_gradient_mad_mean", [Background gradient MAD], [MAD], log-scale:
false) ], ) v(10pt) legend(data) v(10pt) summary-table(data)

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align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[WVF radius-trace follow-up on the fixed fluorescence image set.] linebreak() text(fill: black70, size:
6.8pt)[Garnet trace points sweep WVF across  $(r, d)$  =
(3, 5), (5, 9), (9, 11), (15, 11), (25, 11), (50, 11). Thin horizontal lines are the fixed baseline methods.
Outlined WVF points beat the best fixed baseline on that metric.] ] v(8pt) grid( columns: 2, column-
gutter: 10pt, row-gutter: 10pt, [ wvf-trace-chart(data, "white_noise_gain", [Analytical WNG versus
WVF radius], [log10 WNG], log-scale: true) ], [ wvf-trace-chart(data,
"background_gradient_mad_mean", [Background gradient MAD versus WVF radius], [MAD], log-
scale: false) ], [ wvf-trace-chart(data, "background_gradient_median_mean", [Background gradient
median versus WVF radius], [Median], log-scale: false) ], ) v(8pt) legend(data)

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